# PHYSICAL CHEMISTRY

## GUIDED REVISION PLAN_SCORE ADVANCED [SOLUTION]

## Topic : MOLE CONCEPT & REDOX (Question Bank # 05)

### SINGLE CORRECT

**1. (B)**

**Sol.** $Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O \xrightarrow{0.05\,M\,HCl}$

10 ml, 0.1 M

[Acid base titration - case of non-redox reaction]

**(I) At phenol phthalein end point :** Only $Na_2CO_3$ is converted in $NaHCO_3$.

Balanced chemical equation :

$Na_2CO_2 \cdot NaHCO_3 \cdot 2H_2O + HCl \longrightarrow 2NaHCO_3 + 2H_2O + NaCl$

10ml, 0.1M 0.05M

x ml

→ Since molar ratio is 1 : 1

$\therefore\ 10 \times 0.1 = x \times 0.05 \Rightarrow \boxed{x = 20\,ml}$

**(II) Methyl orange end point :** All the $NaHCO_3$ is converted in $H_2CO_3$

$Na_2CO_3 \cdot NaHCO_3 + 3HCl \longrightarrow 3H_2CO_3 + 3NaCl$

0.1M, 10ml 0.05M

y ml

→ Since molar ratio is 1 : 3

$\therefore\ \dfrac{10 \times 0.1}{0.05 \times y} = \dfrac{1}{3} = \boxed{y = 60\,ml}$

→ Hence $y - x = 60 - 20 = 40$ ml

**2. (B)**

**Sol.** (i) $H_2\overset{+3}{\underline{C_2}}O_4 + \overset{+7}{\underline{Mn}}O_4^- \longrightarrow \overset{+4}{\underline{CO_2}} + Mn^{+2}$

$n = 2$ $n = 5$

100 ml

(ii) $\overset{+7}{\underline{Mn}}O_4^- + I^- \longrightarrow Mn^{+2} + I_2$

10 ml 50 ml

$n = 5$ 1N

From equation (ii) : m. eq. of $MnO_4^-$ = m. eq. of $I^-$

$\Rightarrow M \times 5 \times 10 = 1 \times 50$

$\Rightarrow \boxed{M_{MnO_4^-} = 1M}$

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From equation (i) :

$$\text{m. eq. of } H_2C_2O_4 = \text{m. eq. of } MnO_4^-$$

$$\Rightarrow x \times 2 = 1 \times 100 \times 5$$

$$\Rightarrow x = 250 \text{ m. mol} = \frac{1}{4} \text{ mol}$$

$$\therefore W_{H_2C_2O_4} = \frac{1}{4} \times 90\,g$$

$$= 22.5 \text{ g}$$

**3. (D)**

**Sol.** $\boxed{2M\,AlCl_3 + 3m\ SnCl_4}$

Sincle $[\text{Molarity}]_{Cl^-} = [\text{Molality}]_{Cl^-}$

$$\Rightarrow \frac{n_{Cl^-}}{V_{solution}} \text{in L} = \frac{n_{Cl^-}}{W_{solvent}} \text{in kg}$$

$$\Rightarrow \boxed{V_{solution} \text{ in L} = W_{solvent} \text{in kg}}$$

→ Consider 1 L solution [that means 1 kg solvent]

therefore : $n_{AlCl_4} = 2\,mol \Rightarrow W_{AlCl_3} = 267\,g$

& $n_{SnCl_4} = 3mol \Rightarrow W_{SnCl_4} = 783\,g$

Therefore $W_{solution} = W_{AlCl_3} + W_{SnCl_4} + W_{solvent}$

$= 267 + 783 + 1000$

$= 2050$ g

Now:

$$\text{density of Soln} = \frac{W_{solution}}{V_{solution}} = \frac{2050\,g}{1000\,ml} = 2.05\,g/ml$$

**4. (B)**

**Sol.** $\boxed{\begin{matrix}150\,ml \\ I_2 : 'x'\end{matrix}}$ M

Part-I : $I_2 + Na_2\overset{+2}{\underline{S_2}}O_3 \longrightarrow NaI + Na_2\overset{+2.5}{\underline{S_4}}O_6$

$n = 2 \quad n = 1$

'x' M 15 ml, 0.4 m

V ml

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→ Acc. to law of equivalence :

m. eq. of $I_2$ = m. eq. of $Na_2S_2O_3$

$\Rightarrow V \times x \times 2 = 15 \times 0.4 \times 1$

$\Rightarrow \boxed{V \times x = 3}$ ....(1)

Part-II :

(A) $3I_2 + \overset{\ominus}{6OH} \longrightarrow 5I^- + IO_3^-$ [Balanced]

(150-V)ml (excess)

'x' M 100 ml, 0.3 M

= 30 m. mol

= 24 m. mol (consumed)

[Remaining]

(B) $2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$

6m mol 10 ml

0.3 M

= 3 m. mol

from equation (A) : $\frac{[\text{m.mol}]_{I_2}}{[\text{m.mol}]_{\overset{\ominus}{OH}}} = \frac{1}{2}$

$\Rightarrow (150 - V) \times x = \frac{1}{2} \times 24 = 12$

$\Rightarrow 150x - Vx = 12 \Rightarrow 150x = 15$

$\Rightarrow \boxed{x = 0.1M}$

**5. (A)**

**Sol.** $\boxed{458\,g \text{ Sample } Mn_3O_4}$

(A) (I) $Mn_3O_4 \xrightarrow{F^-} 3Mn^{+2}$

1 mol 3 mol

= 229 g

(II) $Mn^{2+} + K\underset{}{\overset{+7}{Mn}}O_4 \longrightarrow Mn^{+3}$

n = 1 n = 4

3 eq 3 L

3 mol $\frac{1}{4}$M

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(B) $K\overset{+7}{\underline{Mn}}O_4 + H_2C_2O_4 \longrightarrow Mn^{+2} + CO_2$

1.25 N

n = 5

$\text{Molarity} = \frac{1}{4}M$

$\rightarrow$ therefore :

$$\%\text{ purity in the mixture} = \frac{W_{Mn_3O_4}}{W_{Sample}} \times 100$$

$$= \frac{229}{458} \times 100$$

**6. (C)**

**Sol.**

20 ml
$H_2O_2$ : 'x' M

$\xrightarrow[100\text{ ml}]{\text{diluted to}}$ $\therefore$ Molarity is $\boxed{\frac{x}{5}M \; H_2O_2}$

reduced to $\frac{1}{5}$

50 ml soln is taken

$\rightarrow$ $H_2\overset{-1}{\underline{O_2}} + K\overset{+7}{\underline{Mn}}O_4 \xrightarrow{\overset{\oplus}{H}} \overset{0}{\underline{O_2}} + Mn^{+2}$

n = 2 n = 5

$\frac{x}{5}M$ 0.02 M

50 ml 50 ml

$\rightarrow$ Acc. to law of equivalence

$$\frac{x}{5} \times 50 \times 2 = 50 \times 0.02 \times 5$$

$$\Rightarrow \boxed{x = \frac{1}{4}M}$$

$\rightarrow$ Now volume strength of $H_2O_2$ solution
(at 1 atm, 273 k)
= 11.2 × Molarity = 2.8 V

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**7. (B)**

**Sol.** $\boxed{NaHC_2O_4\,(x\text{ mol}) + KHC_2O_4\cdot H_2C_2O_4\,(y\text{ mol})}$

**Case-I: Reaction with $KMnO_4$ : [Redox reaction]**

[Reacting species is $C_2O_4^{-2}$]

$$\overset{+3}{\underline{C}}_2\,O_4^{-2} + K\overset{+7}{\underline{M}}nO_4 \longrightarrow \overset{+4}{\underline{C}}O_2 + Mn^{+2}$$

$(x+2y)$ mol $\quad$ $n = 5$

$n = 2$ $\quad$ V, 0.2 N

$\rightarrow$ Law of equivalence : $(x + 2y) \times 2 = 0.2 \times V$

$\Rightarrow \boxed{V = 10(x+2y)}$ ...(i)

**Case-II: Reaction with 0.12 N NaOH : (Non-redox reaction)**

[Acid-base reaction]

$$\overset{\oplus}{H} + \overset{\ominus}{O}H \longrightarrow H_2O$$

$(x+3y)$ mol $\quad$ 0.12 N

$n = 1$ $\quad$ V

$\Rightarrow (x+3y) \times 1 = 0.12 \times V$

$\Rightarrow \boxed{V = \frac{100}{12}\times[x+3y]}$ ....(ii)

(i) / (ii)

$\boxed{\frac{x}{y} = \frac{3}{1}}$

**8. (D)**

**Sol.** **Kjeldahl method :**

$$\underset{1.4\,g}{\boxed{O.C.}} \xrightarrow{\text{evolved } NH_3} NH_3$$

(I) $2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2\,SO_4$

10m. mol $\quad$ 60ml, $\frac{M}{10}$

= 6 m. mol [Initial]

= 5 m. mol [Consumed in the reaction].

Remaining

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(II) $H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$

1 m. mol 20 ml

$\frac{M}{10}$

$= 2$ m. mol

therefore m. mol of $NH_3$ produced $= 10$

$\Rightarrow$ Since m. mol of N = m. mol of $NH_3$

$\Rightarrow$ m. mol of N = 10

$\Rightarrow$ mol of $N = \frac{10}{1000}$

$\Rightarrow W_N = \frac{1}{100} \times 14\,g$

$$\text{therefore } \%\,N = \frac{W_N}{W_{O.C.}} \times 100 = \frac{\frac{14}{100}}{1.4} \times 100$$

$= 10\%$

**9. (A)**

**Sol.** $\boxed{FeO\,(a\,mol) + Fe_2O_3\,(b\,mol)}$

Case-I : $KMnO_4$ reacts only with FeO, since it is an oxidizing agent.

$\rightarrow K\overset{+7}{\underline{M}}nO_4 + \overset{+2}{\underline{Fe}}O \longrightarrow Fe^{+3} + Mn^{+2}$

$\frac{2}{5}$ M, $\boxed{n = 5}$ a mol $n = 1$

0.1 L $n = 1$

* $\boxed{\text{Eq. of FeO} = \text{Eq. of } KMnO_4}$ = eq. of $Fe^{+3}$

$\Rightarrow a \times 1 = \frac{2}{5} \times 0.1 \times 5 \Rightarrow \boxed{a = \frac{2}{10}\,mol}$

Case-II: Now, Zn reduced all the $Fe^{+3}$ in $Fe^{+2}$

$\rightarrow$ Total amount of $Fe^{+3}$ in mix $= (a + 2b)$ mol

{'a' mol is obtained from the oxidation of $Fe^{+2}$ and '2b' mol of $Fe^{+3}$ from $Fe_2O_3$}

$Zn + Fe^{+3} \longrightarrow Zn^{+2} + Fe^{+2}$

(a+2b) mol –

– (a+2b) mol

[POAC]

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Case-III: $Fe^{+2} + \overset{+6}{K_2Cr_2O_7} \longrightarrow Fe^{+3} + Cr^{+3}$

$n = 1 \qquad n = 6$

$(a+2b)\text{mol} \quad \frac{2}{15}M_1\,1L$

$= (a+2b)\times 1 = \frac{2}{15}\times 1\times 6 \Rightarrow b = \frac{3}{10}\text{mol}$

To calculate $\frac{a}{b} \Rightarrow \frac{a}{b} = \frac{\frac{2}{10}}{\frac{3}{10}} = \frac{2}{3}$

## MULTIPLE CORRECT

**10. (B,C)**

**Sol.** Let volume of $NH_{3(g)} = V$ ml

Dissociation of $NH_{3(g)}$ :

$$2NH_{3(g)} \longrightarrow N_{2(g)} + 3H_{2(g)}$$

$t = 0 \quad V \quad - \quad -$

$t \quad V - V_1 \quad \frac{V_1}{2} \quad \frac{3V_1}{2}$

→ After dissociation of $NH_3$, volume becomes 90 ml. [$NH_3$ is not completely dissociated, since after reaction with $H_2SO_4$ volume is converted in 80 ml. $H_2SO_4$ reacts with $NH_3$ completely to get $(NH_4)_2SO_4$

→ therefore after reaction:

* $V_{N_2} + V_{H_2} = 80$

$\Rightarrow \frac{V_1}{2} + \frac{3V_1}{2} = 80 \Rightarrow \boxed{V_1 = 40\,\text{ml}}$

→ and remaining vol. of $NH_3$ :

$V_{NH_3} = V - V_1 = 10$

$\Rightarrow \boxed{V = 50\,\text{ml}}$

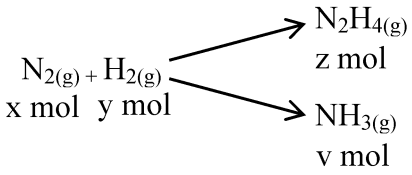
→ that mean 80% $NH_3$ is dissociated in $N_2$ and $H_2$.

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**11. (A,B,D)**

**Sol.**




Chemical reactions are

Step-I) $N_{2(g)} + 2H_{2(g)} \longrightarrow N_2H_{4(g)}$

Step-II) $N_2H_{4(g)} + H_{2(g)} \longrightarrow 2NH_{3(g)}$

→ Check first option :

(A) (2, 2, 1, 0)

(I) $N_{2(g)} + \overset{(LR)}{2H_{2(g)}} \longrightarrow N_2H_{4(g)}$

| | | |
|---|---|---|
| 2mol | 2mol | – |
| 1mol | – | 1mol |

Next step is possible only if $H_2$ is in excess, but since it is LR, therefore $n_{NH_3} = 0$

First option is correct.

(B) (3, 8, 1, 4)

(I) $\overset{(LR)}{N_{2(g)}} + \overset{(Excess)}{2H_{2(g)}} \longrightarrow N_2H_{4(g)}$

| | | |
|---|---|---|
| 3mol | 8mol | – |
| – | 2 mol | 3 mol |

II reaction is possible.

(II) $N_2H_{4(g)} + \overset{(L.R)}{H_{2(g)}} \longrightarrow 2NH_{3(g)}$

| | | |
|---|---|---|
| 3 mol | 2 mol | – |
| 1 mol | – | 4mol |

Final cond. $N_2H_4$ : 1 mol

$NH_3$ : 4 mol

→ Second option is also correct.

→ Similarly check other options.

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**12. (A)**

**Sol.** * 1 L water sample is taken.

* Heavily polluted means, concentration of dissolved oxygen ($O_2$) in water is less than 6.4 mg/L

* Mass of dissolved $O_2$ in 1L sample

$$= 6.4\frac{mg}{L} \times 1L$$

$$= 6.4\ mg$$

* m. mol of dissolved $O_2 = \frac{6.4}{32} = 0.2$

$\rightarrow$ from the equation :

(I) $Mn(OH)_2 + \frac{1}{2}O_2 + H_2O \longrightarrow 2Mn(OH)_{3(s)}$

0.2 m.mol –

– 0.8 m. mol

(II) $2Mn(OH)_3 + 2KI + 3H_2SO_4 \longrightarrow 2MnSO_4 + k_2SO_4 + I_2 + 6H_2O$

0.8 m. mol –

– 0.4 m. mol

(III) $I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$

0.4 m. mol –

– 0.8 m. mol

$\rightarrow$ therefore m. mol of $Na_2S_2O_3$ required = 0.8

$\Rightarrow$ M × V = 0.8

$\Rightarrow$ 0.01 × V = 0.8

$\Rightarrow$ $\boxed{V = 80\,ml}$

**13. (C)**

**Sol.** 1L water sample is taken :

$\rightarrow$ concentration of $O_2$ required for optimum fit for aquatic life = 9.6 mg/L

$\rightarrow \therefore$ m.mol of $O_2 = \frac{9.6}{32}\frac{m.mol}{L} \times 1L$

$= 0.3$ m. mol

$\rightarrow$ from the equation :

(I) $2Mn(OH)_2 + \frac{1}{2}O_2 + H_2O \longrightarrow 2Mn(OH)_{3(s)}$

0.3 m. mol –

– 1.2 m. mol

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(II) $2Mn(OH)_{3(s)} + 2KI + 3H_2SO_4 \longrightarrow 2MnSO_4 + K_2SO_4 + I_2 + 6H_2O$

1.2 m. mol – – 0.6 m. mol

therefore minimum m. mol of $Na_2S_2O_3$ required = 1.2

$\Rightarrow$ M × V = 1.2

$\Rightarrow$ 0.1 × V = 1.2

$\Rightarrow$ $\boxed{V = 12\,ml}$

**14. (B)**

**Sol.** (B) is the correct statement.

(A) Incorrect : more dissolved $O_2$ in water requires more moles of $Na_2S_2O_3$

(C) Incorrect : $n_{Mn(OH)_2}$ and $n_{Mn(OH)_3}$ will be same (according to reaction stoichiometry)

(D) Incorrect : 1 mol of dissolved $O_2$ will require 4 mol of $Na_2S_2O_3$ for estimation.

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## SECTION-II [INTEGER TYPE]

**1. (4)**

**Sol.** $[\text{density}]_{(C_2H_5)_4Pb} = 6.48\,g/ml,\ vol = 2ml$

$\therefore$ Mass of $(C_2H_5)_4Pb$ reqd. for 1L gasoline

= v × d = 2 ml × 6.48 g/ml

= 2 × 6.48 g

$\therefore$ for 25 L gasoline = 2 × 6.48 × 25 g

From the given equation:

$4\,C_2H_5Cl + 4\,Na - Pb \longrightarrow (C_2H_5)_4Pb + 4\,NaCl + 3\,Pb$

– $\frac{2\times6.48\times25}{324}$ mol

4 mol = 1 mol

Ans. 4 mol $C_2H_5Cl$

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**2.** **(1)**

**Sol.** $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24\,H_2O$

Given: $W_0 = 5000\ g$

$$n_0 = \frac{5000}{16}\,mol$$

Since molar ratio of 'S' and 'O' is $\frac{1}{10}$

therefore: $\frac{n_S}{n_O} = \frac{1}{10}$

$$\Rightarrow \quad n_S = \frac{1}{10} \times n_O$$

$$= \frac{1}{10} \times \frac{5000}{16}\,mol$$

therefore: $W_S = \frac{5000}{10 \times 16} \times 32g$

$= 1\ kg$

**3.** **(5)**

**Sol.** $\underset{n_2}{Br_2} + \overset{\ominus}{OH} \longrightarrow \underset{n_1 = 2}{Br^-} + \text{Oxidized product}$ ($+2e^-$)

→ D is proportionation reaction :

* n - factor of $Br_2 = \frac{n_1 n_2}{n_1 + n_2}$

→ Given eq. wt. of $Br_2 = 96$

$$\Rightarrow \frac{M_{Br_2}}{n - factor} = 96$$

$$\Rightarrow n - factor = \frac{160}{96} = \frac{5}{3}$$

$$\Rightarrow \frac{n_1 \times n_2}{n_1 + n_2} = \frac{5}{3} \qquad \{ n_1 = 2$$

$$\Rightarrow \boxed{n_2 = 10}$$

To loose 10 mol of $e^-$ & by 1 mol $Br_2$. Oxidation No. of 'Br' in the oxixized product should be '+5'.

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## SECTION-IV [MATRIX MATCH]

**1. (A-Q,R; B-P,S; C-Q,R)**

**Sol.** Two separate solution :

| | |
|---|---|
| $KMnO_4 \xrightarrow[\text{Medium}]{\text{Neutral}} Mn^{+4}$ <br> $n = 3$ | $O_2 \longrightarrow O^{-2}$ <br> $n = 4$ |

*** First experiment :**

(I) $KMnO_4 + \overset{+2}{\underline{Fe}}\overset{+3}{\underline{Cr_2}}O_4 \longrightarrow Mn^{+4} + \overset{+3}{\underline{Fe_2}}O_3 + \underline{K_2}\overset{+6}{Cr}O_4$

$n = 3$

$\frac{1}{20}M,\ 20\,ml \qquad n = 7$

$\rightarrow$ m. eq. of $KMnO_4$ = m. eq. of $FeCr_2O_4$

$\Rightarrow \frac{1}{20} \times 20 \times 3 = m.mol \times 7$

$\Rightarrow (m.mol)_{FeCr_2O_4} = \frac{3}{7}$

(II) $O_2 + FeCr_2O_4 \longrightarrow O^{-2} + Fe_2O_3 + K_2CrO_4$

$n = 4 \qquad n = 7$

$\frac{1}{10}M,\ 10\,ml$

$\rightarrow$ m.eq.of $O_2$ = m.eq.of $FeCr_2O_4$

$\Rightarrow \frac{1}{10} \times 10 \times 4 = m.mol \times 7$

$\Rightarrow [m.mol]_{FeCr_2O_4} = \frac{4}{7}$

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**Experiment-II :**

* n-factor of $CuFeS_2$ :

$$CuFeS_2 + 4H_2O \longrightarrow Cu^{+} + Fe^{+2} + 2SO_2 + 8\overset{\oplus}{H}$$

$\boxed{n = 11}$ $\qquad +11e^{-}$

(I) m. eq. of $KMnO_4$ = m. eq. of $CuFeS_2$

$$\Rightarrow \frac{1}{20} \times 20 \times 3 = m.mol \times 11$$

$$\Rightarrow (m.mol)_{CuFeS_2} = \frac{3}{11}$$

(II) m. eq. of $O_2$ = m. eq. of $CuFeS_2$

$$\Rightarrow [m.mol]_{CuFeS_2} = \frac{4}{11}$$

**Experiment-III :**

* n - factor of FeS :

$$FeS + 2H_2O \longrightarrow Fe^{+3} + SO_2 + 4\overset{\oplus}{H} + 7e^{-}$$

$\boxed{n = 7}$

(I) $\boxed{m.eq.of\ KMnO_4 = m.eq.of\ FeS}$

(II) $\boxed{m.eq.of\ O_2 = m.\ eq.of\ FeS}$

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